



II. NUCLEI ON THE AGB:

Asymptotic Giant Branch

From light element nucleosynthesis to the *s*- and *i*-processes

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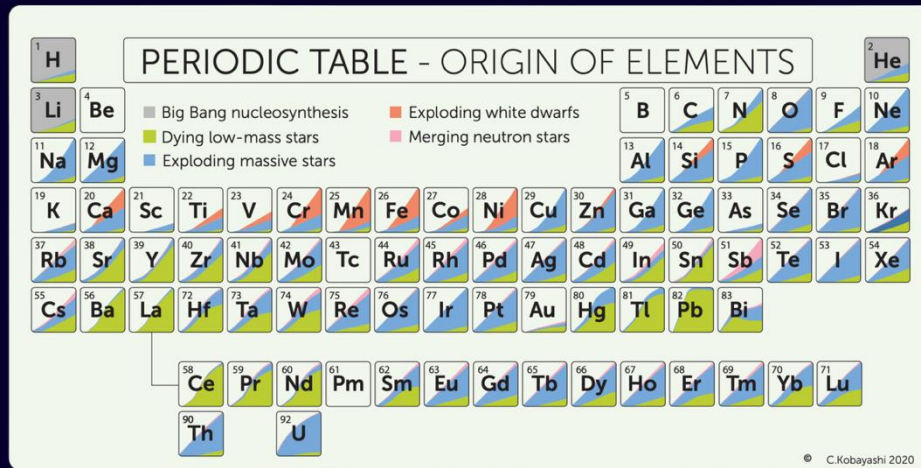
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STELLAR NUCLEOSYNTHESIS
2026 SCHOOL, UNIVERSITY OF HERTFORDSHIRE

AGB Stars Matter For Chemical Evolution

Kobayashi, Karakas & Lugaro 2020 (ApJ)



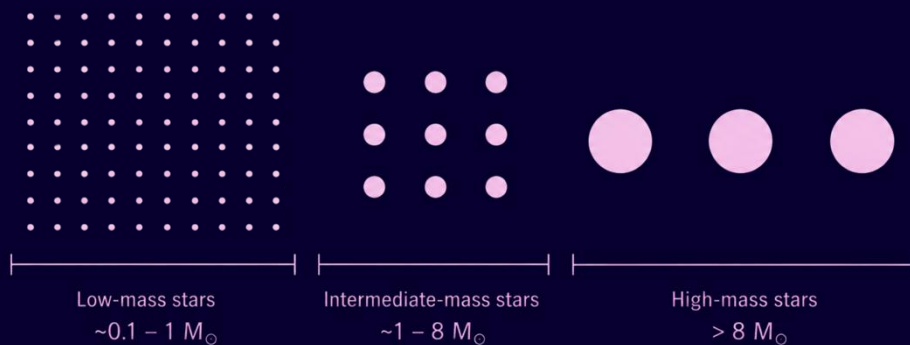
ABUNDANT: Evolutionary endpoint for low- and intermediate-mass stars (LIMS).

LONG-LIVED: Extended lifetimes allow repeated cycles of nucleosynthesis and mixing.

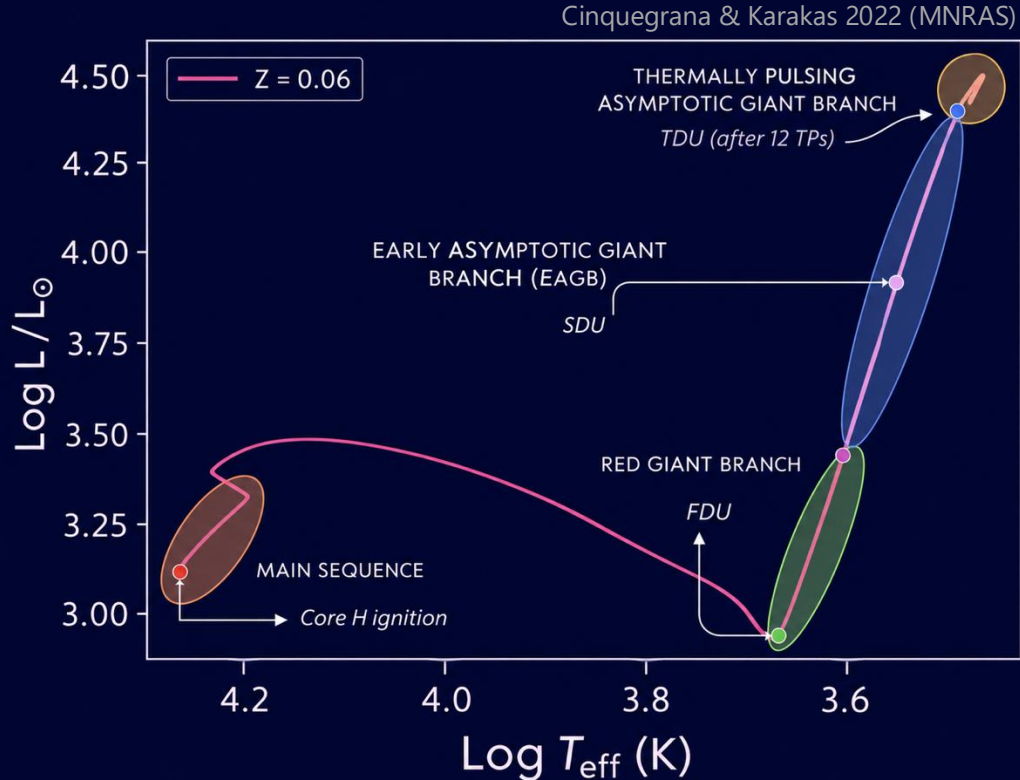
EFFICIENT POLLUTERS: Strong AGB winds return large amounts of processed material to the ISM.

SENSITIVE: Their yields depend strongly on mass and metallicity, producing distinctive abundance signatures.

CHEMICAL ENRICHMENT: LIMS produce significant amounts of C, N, and heavy elements from Sr to Pb through neutron-capture nucleosynthesis.



AGB Structure



POST CORE He BURNING: Star contracts but is not massive enough to ignite C burning. Instead, ignites He burning shell below pre-existing H burning shell.

EARLY AGB: He shell burns progressively itself thinner in mass until it is thermally unstable (the thin shell instability).

THERMAL PULSE: Thermonuclear runaway produces a flash originating from the He-shell. This drives an expansion of the outer layers.

AFTER THE PULSE: He burning subsides and star contracts. Sufficiently massive stars reignite H-burning and enter the TP-AGB.

AGB Structure

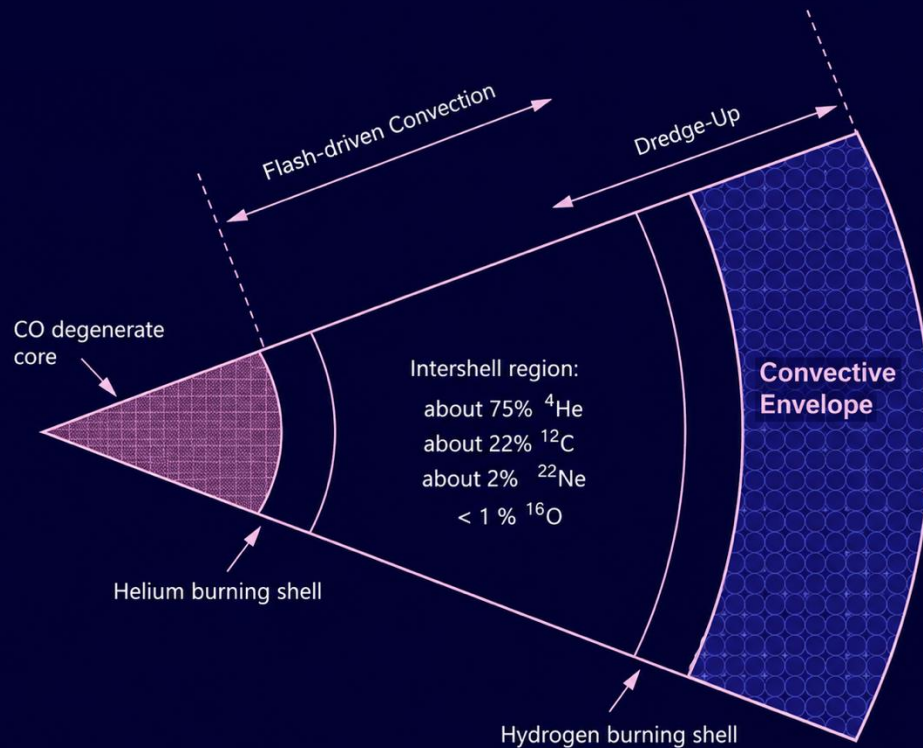


Figure 14 from Karakas & Lattanzio 2014 (Dawes Rev, PASA)

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The Thermal Pulse Cycle

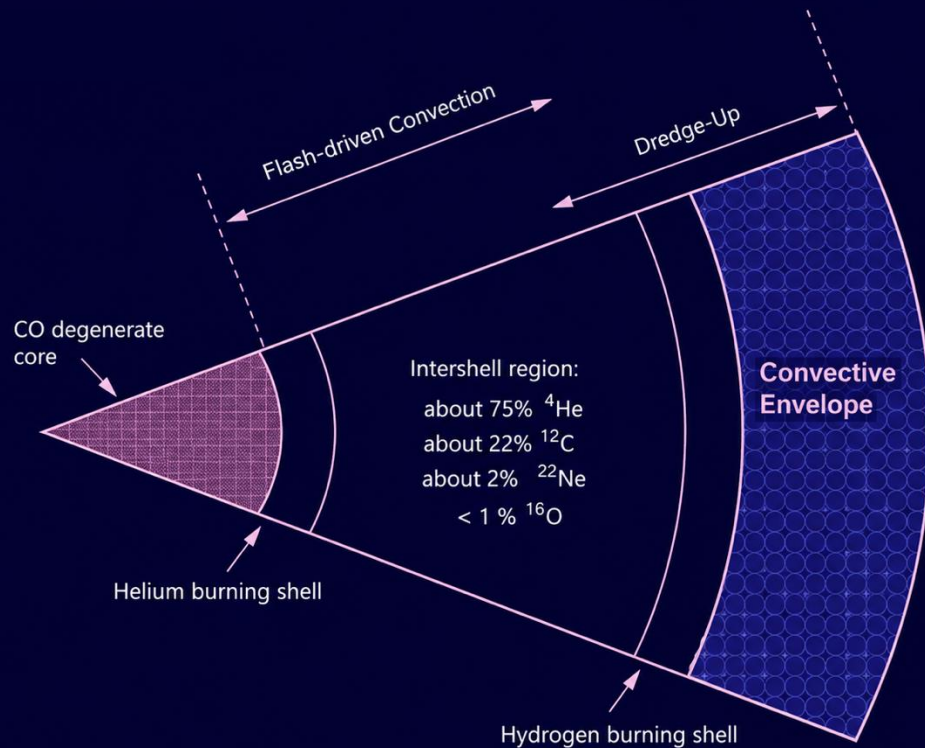


Figure 14 from Karakas & Lattanzio 2014 (Dawes Rev, PASA)

THERMAL PULSE: He-burning shell becomes thermally unstable. A brief flash from the He-shell drives a convective zone through the intershell.

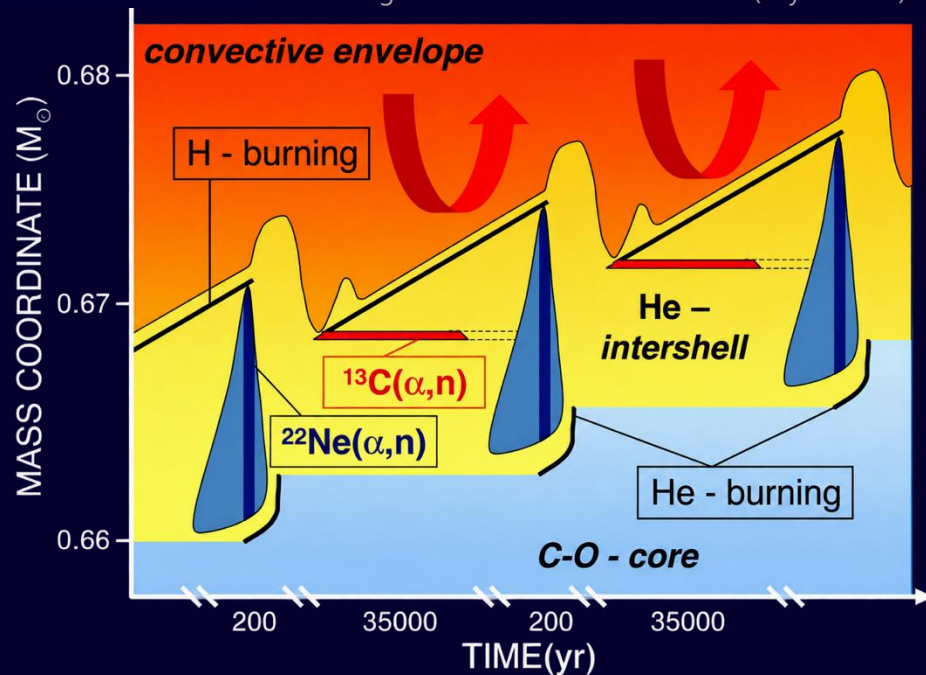
POWER DOWN: He-shell burning rapidly subsides. The star expands and the H-burning shell may be temporarily extinguished.

THIRD DREDGE UP: The convective envelope moves inward into the He-rich intershell, mixing He-burning products and s-process material to the surface.

INTERPULSE: H-burning shell reignites and provides most of the stellar luminosity. Pressure builds up on the He shell until the next thermal pulse.

The Thermal Pulse Cycle & the s-process

Figure 1 from Wallner et al. 2016 (Phys. Rev. C)



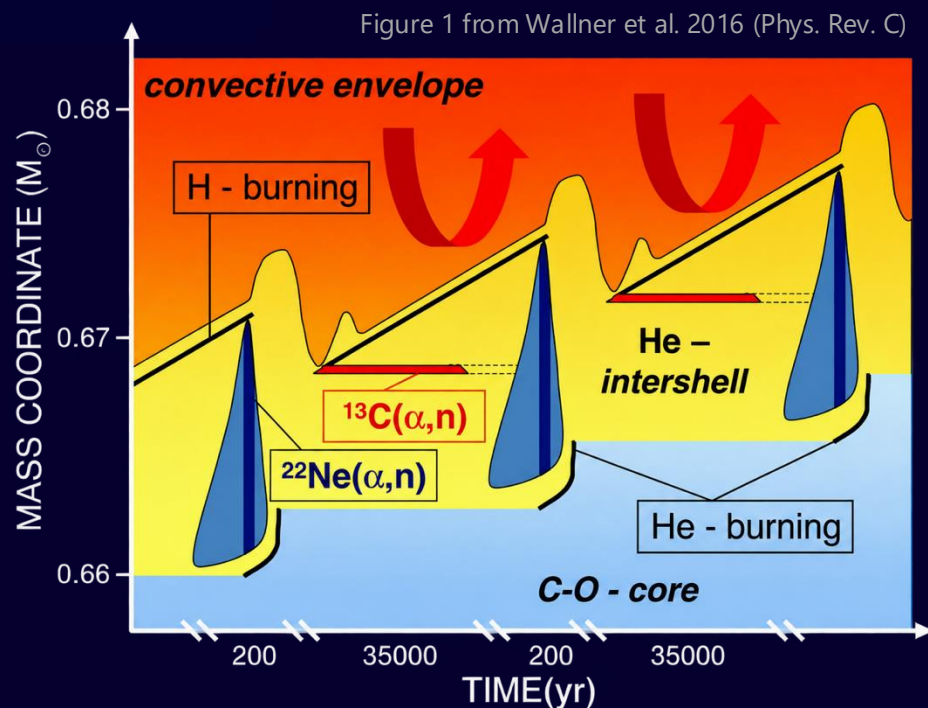
SOURCES OF FREE NEUTRONS:

- $^{13}\text{C} (\alpha, n) ^{16}\text{O}$: Operates at ~ 100 MK, producing low neutron densities over long timescales.
- $^{22}\text{Ne} (\alpha, n) ^{25}\text{Mg}$: Activated at > 300 MK, producing high neutron densities in short bursts.

MAIN SITES OF CAPTURE:

- ^{13}C pocket: Neutrons released radiatively during the interpulse period, providing the largest neutron exposure.
- He-flash convection zone: Neutrons released during thermal pulses, producing a brief, high-density neutron burst.

The Thermal Pulse Cycle



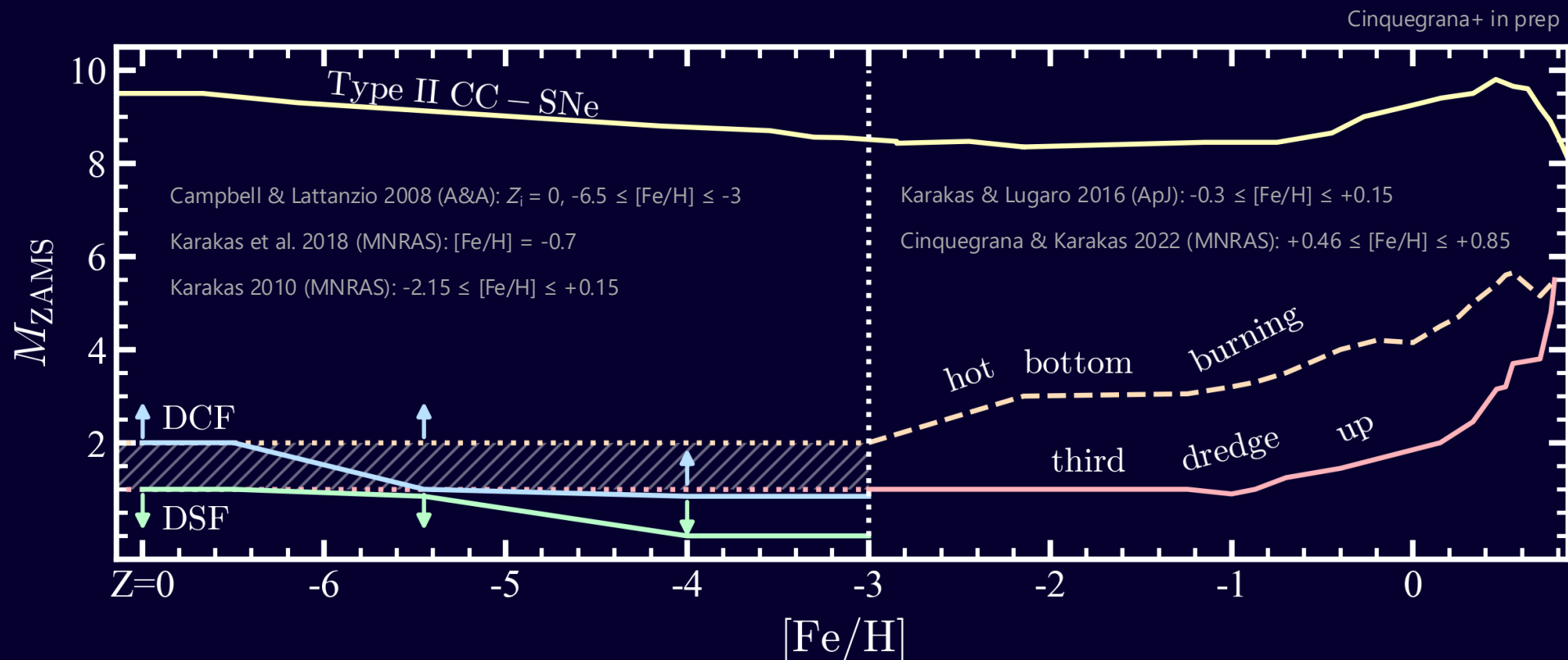
THERMAL PULSE → POWER DOWN → THIRD DREDGE UP → INTERPULSE → REPEAT (?)

CYCLE EVOLVES: Each pulse alters the core, envelope, temperature, composition, and mass-loss rate, changing subsequent nucleosynthesis.

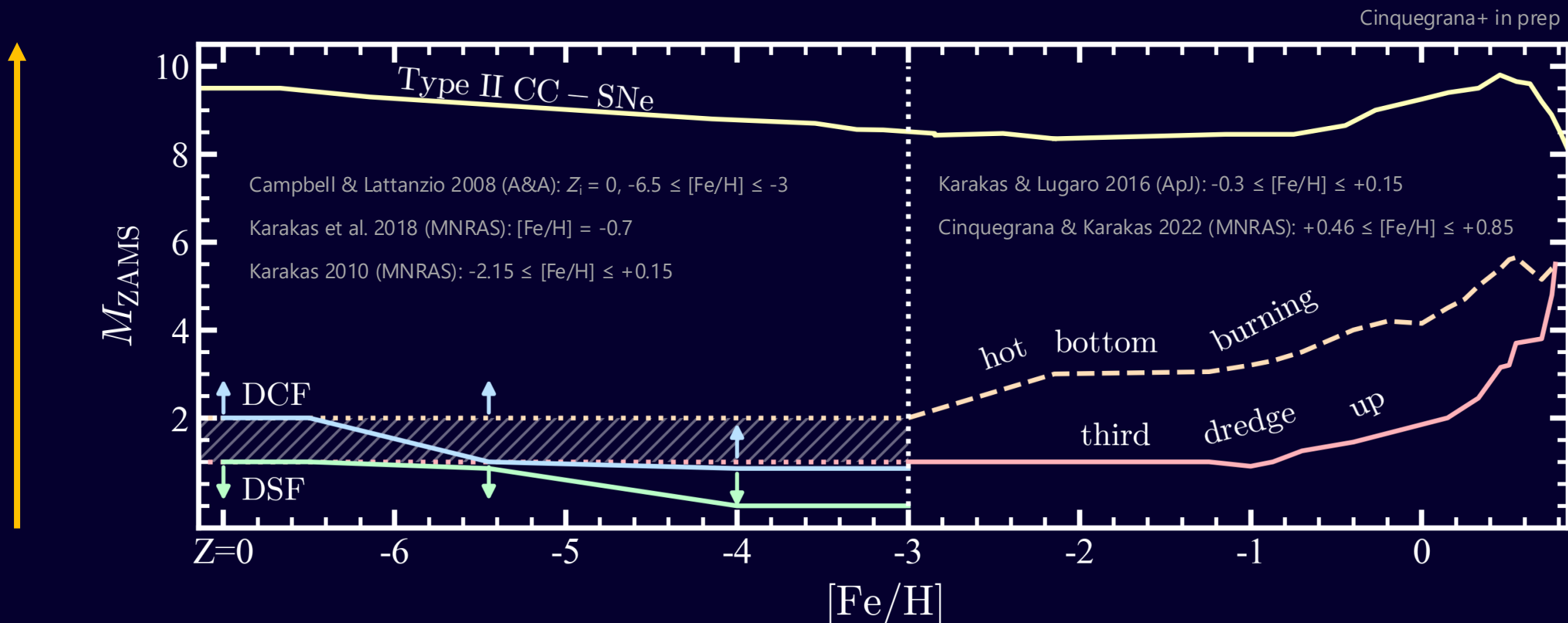
MASS RETURNED TO ISM: Strong stellar winds progressively remove the envelope, ejecting repeatedly processed and mixed material throughout the TP-AGB.

MASS & METALLICITY DEPENDENCE: M_{ZAMS} and Z_i determine the number and strength of pulses, 3DU efficiency, ^{13}C -pocket formation, and HBB activation.

M_{ZAMS} & Z_i vs the Thermal Pulse Cycle



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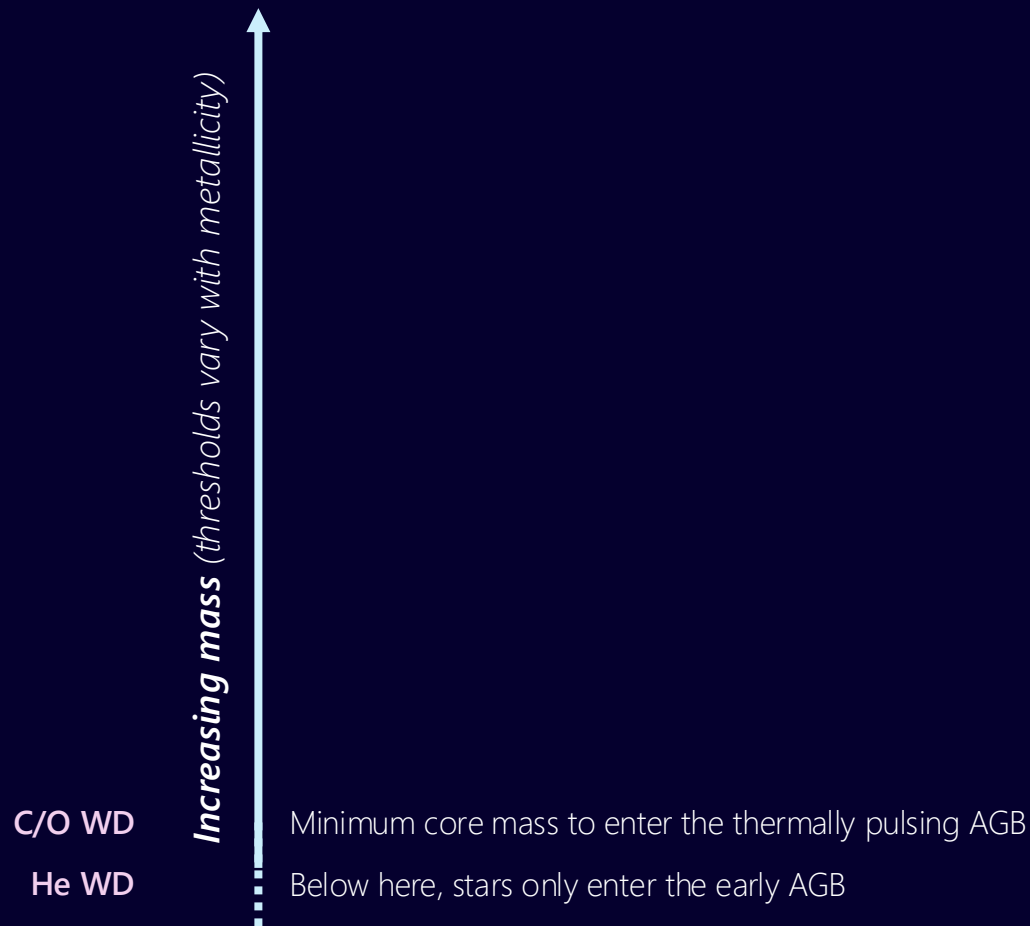


M_{ZAMS} & Z_i vs the Thermal Pulse Cycle

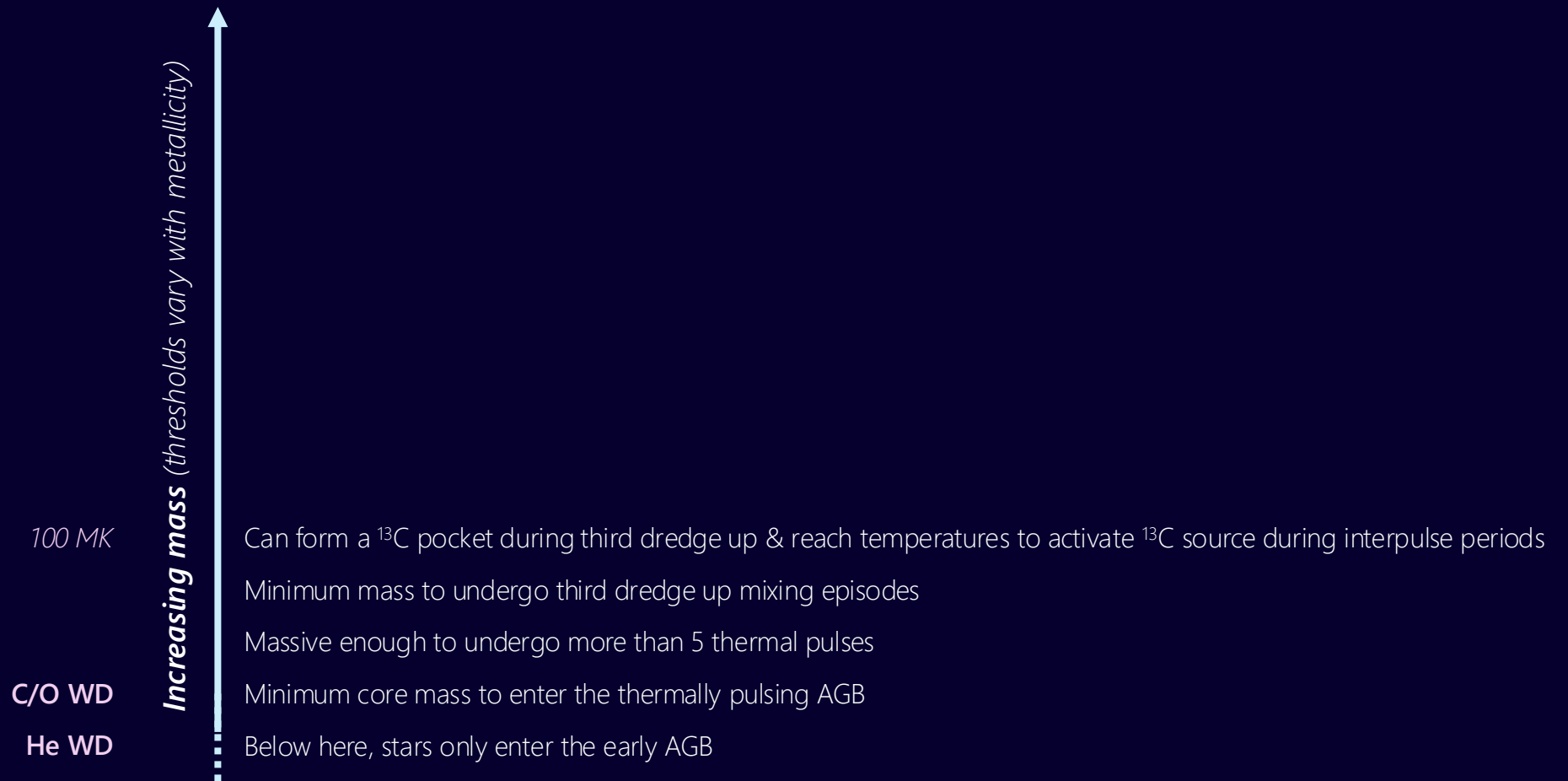
Increasing mass (thresholds vary with metallicity)



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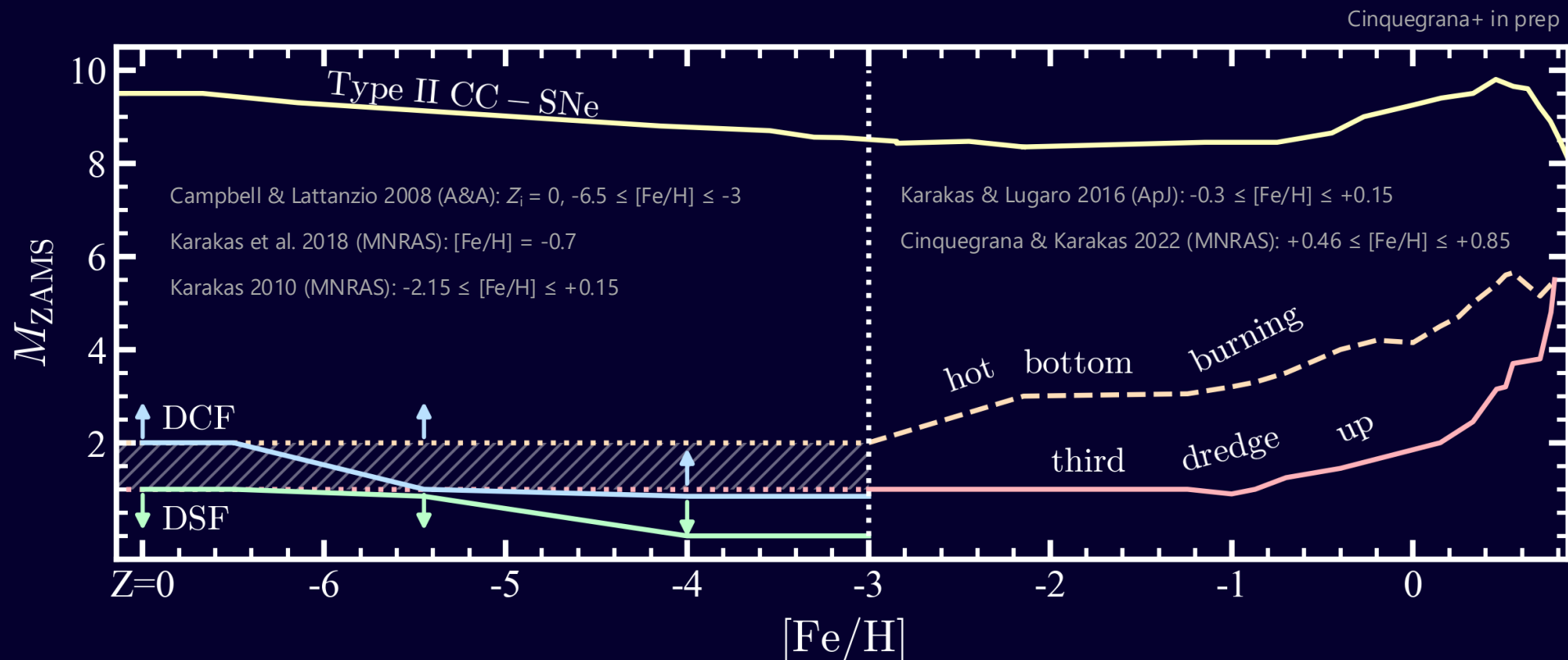


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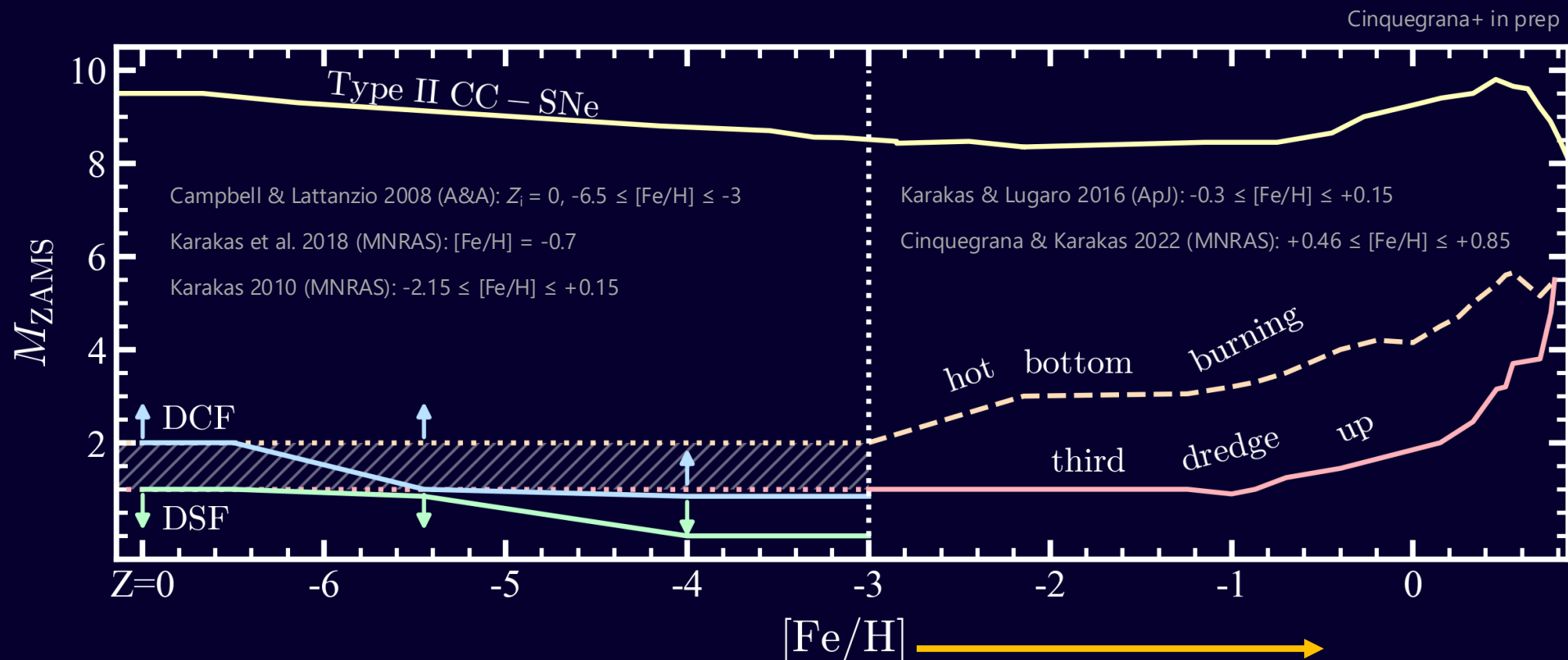
Type II CC-SNe	Type II core collapse supernovae progenitors
ONe WD or EC-SNe	Ignite central carbon on the early AGB [<i>super AGBs</i>]
> 80 MK	Temperatures sustain Mg-Al & Ne-Na chains at the base of the envelope
Hybrid C/O + ONe WD	Ignite carbon off-centre on the early AGB
70-80 MK	Temperatures ... full CNO cycle at the base of the envelope
50 MK	Temperatures ... CN cycle at the base of the envelope
40-50 MK	Temperatures can sustain p-p chains at base of the convective envelope during interpulse periods
40-50 MK	Temperatures during third dredge up are too high to preserve ^{13}C pocket
300 MK	Temperatures reached to activate ^{22}Ne neutron source during thermal pulses
100 MK	Peak neutron exposure from ^{13}C neutron source
	Can form a ^{13}C pocket during third dredge up & reach temperatures to activate ^{13}C source during interpulse periods
	Minimum mass to undergo third dredge up mixing episodes
C/O WD	Massive enough to undergo more than 5 thermal pulses
He WD	Minimum core mass to enter the thermally pulsing AGB
	Below here, stars only enter the early AGB

Increasing mass (thresholds vary with metallicity)

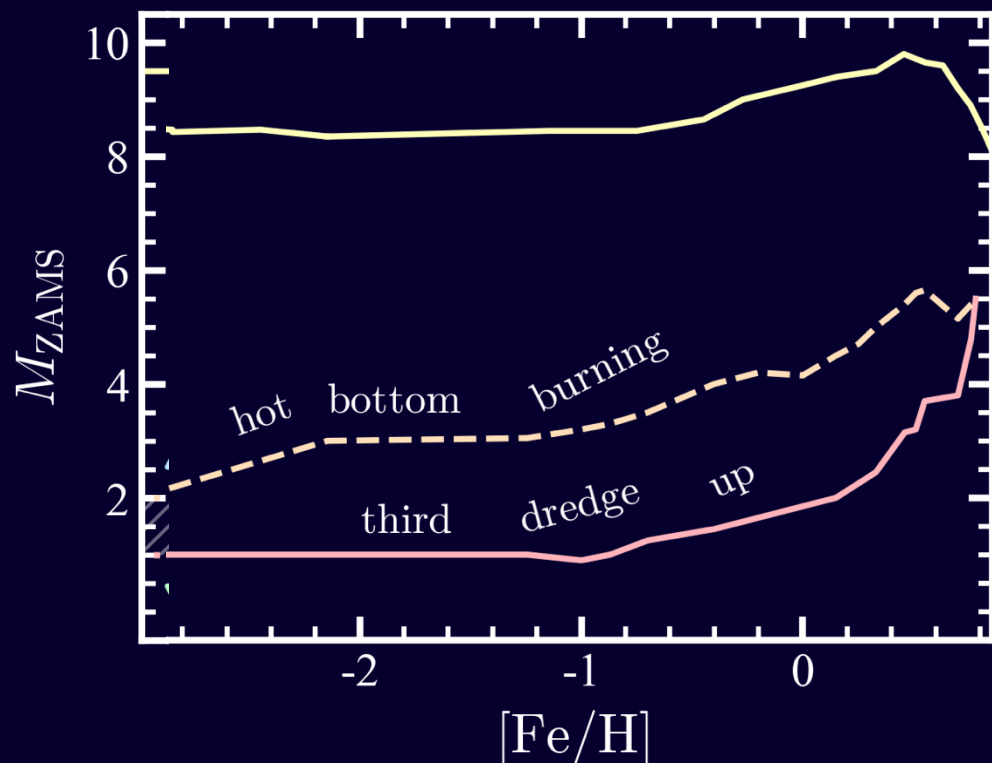
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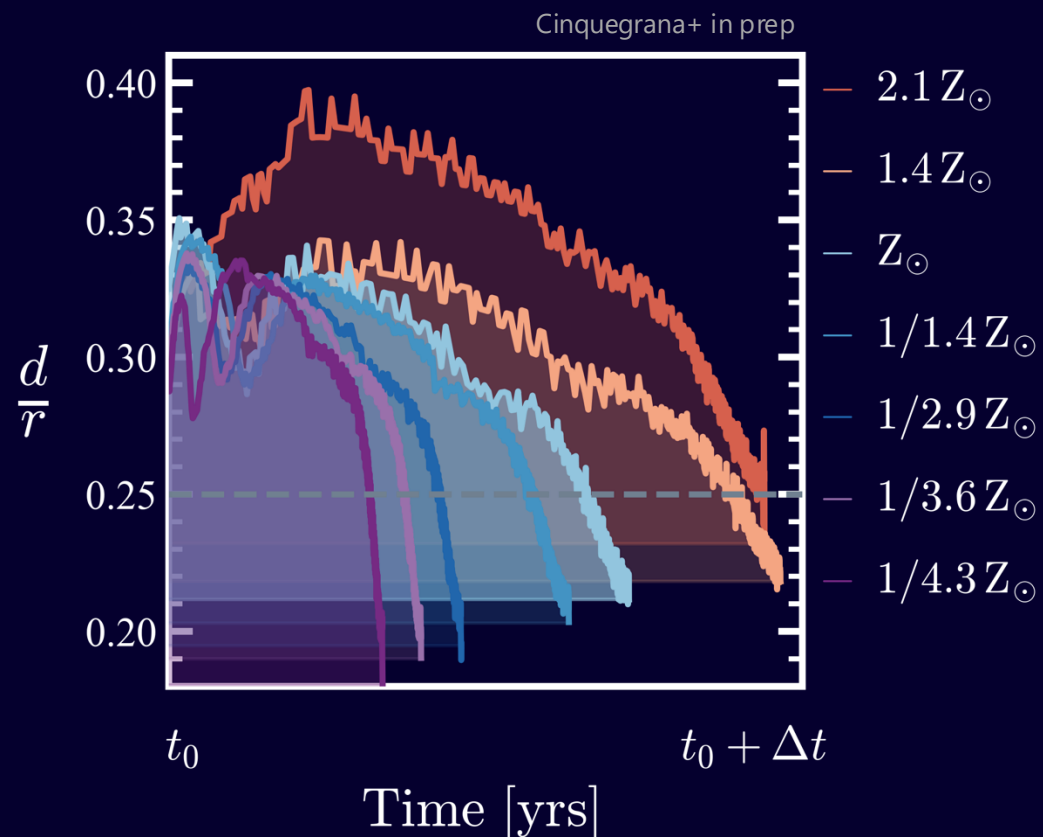
OPACITY (κ): $\uparrow Z_i \rightarrow \uparrow \kappa \rightarrow$ energy escapes less efficiently $\rightarrow$ cooler, less luminous star

MEAN MOLECULAR WEIGHT (μ): $\uparrow Y_i$ with $Z_i \rightarrow \uparrow \mu \rightarrow$ contraction and heating $\rightarrow$ hotter, more luminous star

ENERGY GENERATION RATES (ϵ_{nuc}): Changes with composition and burning mode $\rightarrow$ alters core temperature, density and luminosity

MASS LOSS RATES: $\uparrow Z_i \rightarrow$ stronger mass loss $\rightarrow$ lower stellar mass, changing the subsequent evolution

Z_i & Thermal Pulses

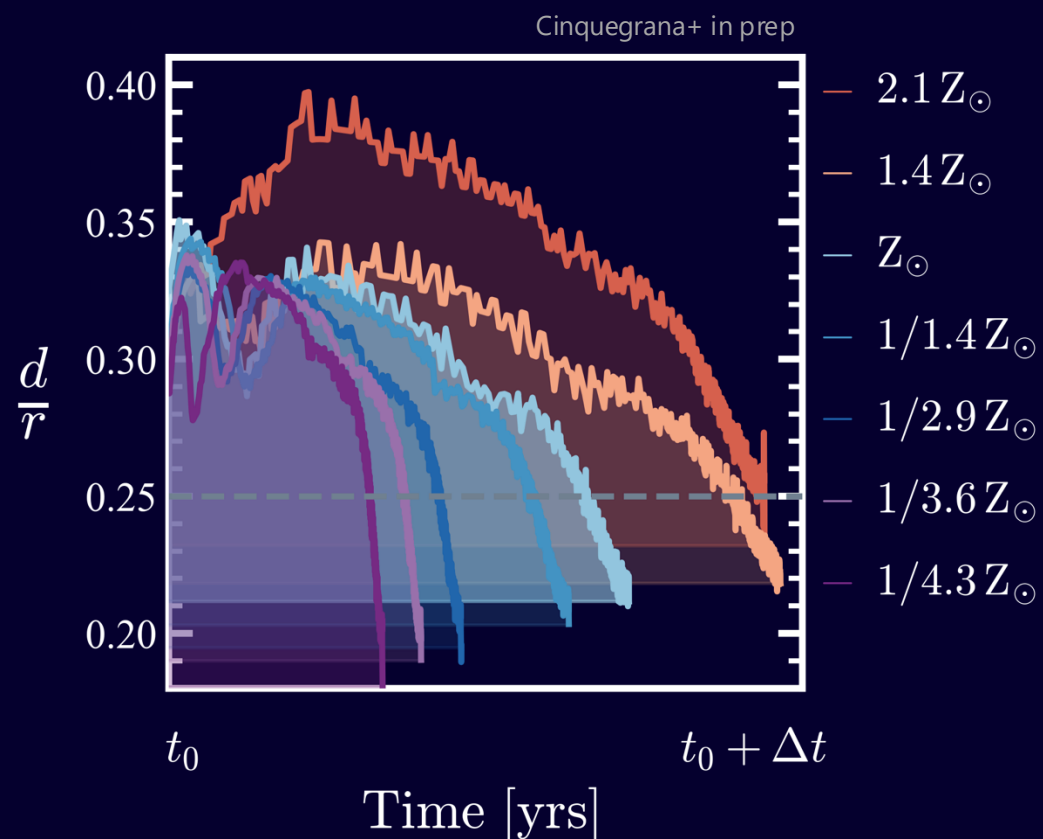


d/r : the He-burning shell's thickness over its outer radius — across the TP-AGB (t_0 to $t_0 + \Delta t$), for models spanning $2.1 Z_{\odot}$ down to $1/4.3 Z_{\odot}$.

STABILITY CRITERION: the shell is thermally unstable (can trigger a pulse) when $d/r < \chi_{\rho}/4$, which for an ideal gas gives $d/r \lesssim 1/4$ — the dashed line marks this threshold.

METALLICITY PATTERN: higher- Z models sustain a thicker shell (higher, longer-lived d/r) across more pulses; lower- Z models thin faster and drop below the threshold earlier in the TP-AGB. lower metallicity means lower opacity and a more compact, higher-density core/shell structure, so the He shell burns down and thins relative to its radius more rapidly, driving d/r below $1/4$ sooner.

Z_i & Third Dredge Up

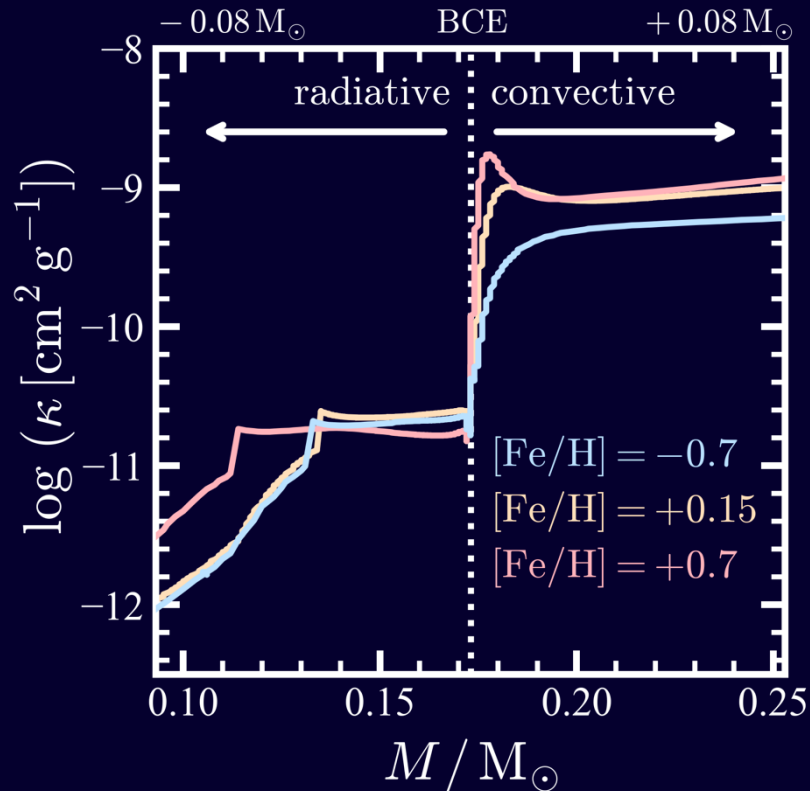


METALLICITY CONTROLS TDU: He-burning shell becomes thermally unstable. A brief flash from the He-shell drives a convective zone through the intershell.

CARBON ENRICHMENT: At low Z_i , less initial oxygen means less carbon is needed to reach $C/O > 1$.

DOUBLE EFFECT: Low metallicity both enhances dredge-up and makes carbon-star formation easier; at high Z , TDU is weaker while more carbon is required.

Z_i & Hot Bottom Burning



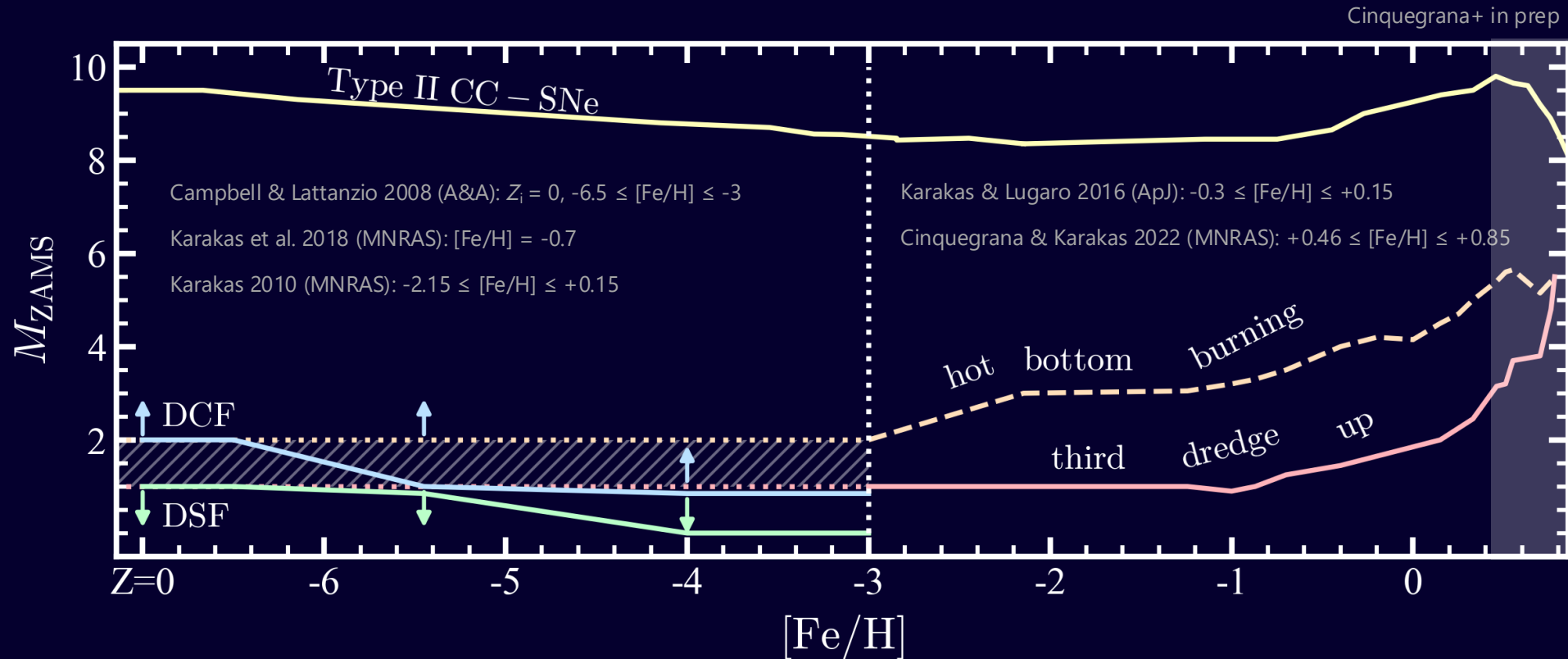
Cinquegrana+ in prep

HIGHER $Z_i \rightarrow$ COOLER HBB: At fixed initial mass, the maximum temperature at the base of the convective envelope generally decreases with increasing metallicity.

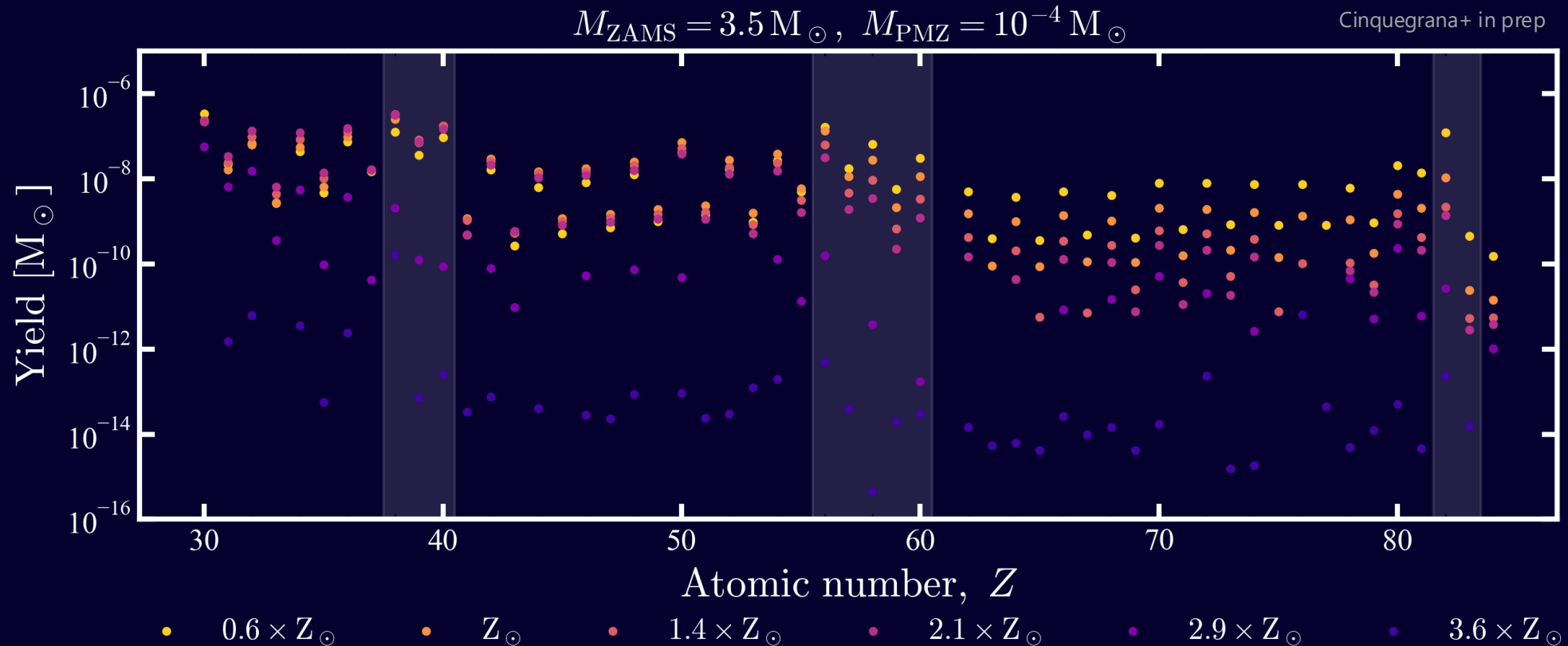
MASS THRESHOLD SHIFTS: HBB therefore requires higher initial masses at higher Z , with the hottest, most massive models reaching the strongest H-burning conditions.

NUCLEOSYNTHESIS CHANGES: As T_{BCE} increases, burning at the base of the convective envelope progresses from CN cycling (≥ 50 MK) toward full CNO and Ne–Na/Mg–Al cycling ($\geq$ MK).

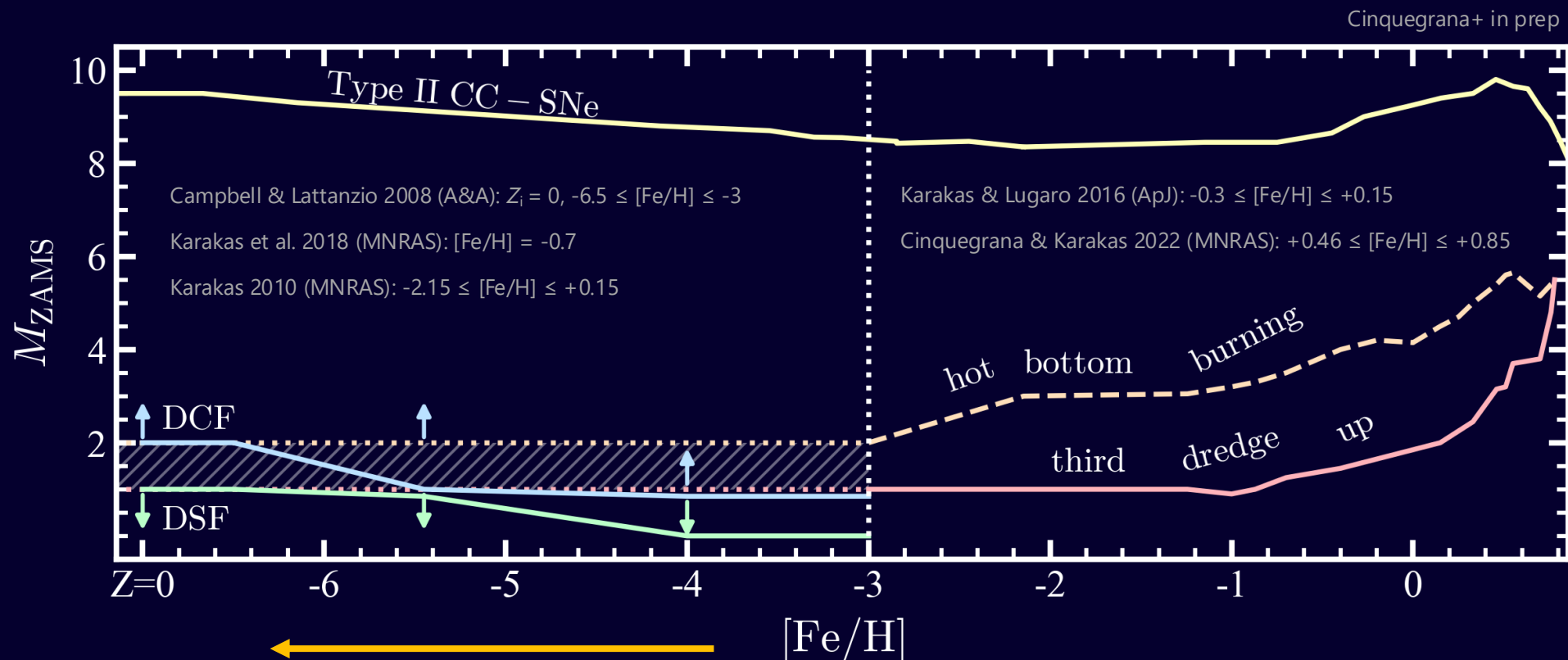
Z_i & Opacity vs Mean Molecular Weight



Z_i & Slow Neutron Capture Processes



M_{ZAMS} & Z_i vs the Thermal Pulse Cycle



Proton Ingestion Episodes (PIEs)

Campbell & Lattanzio (2008, A&A)

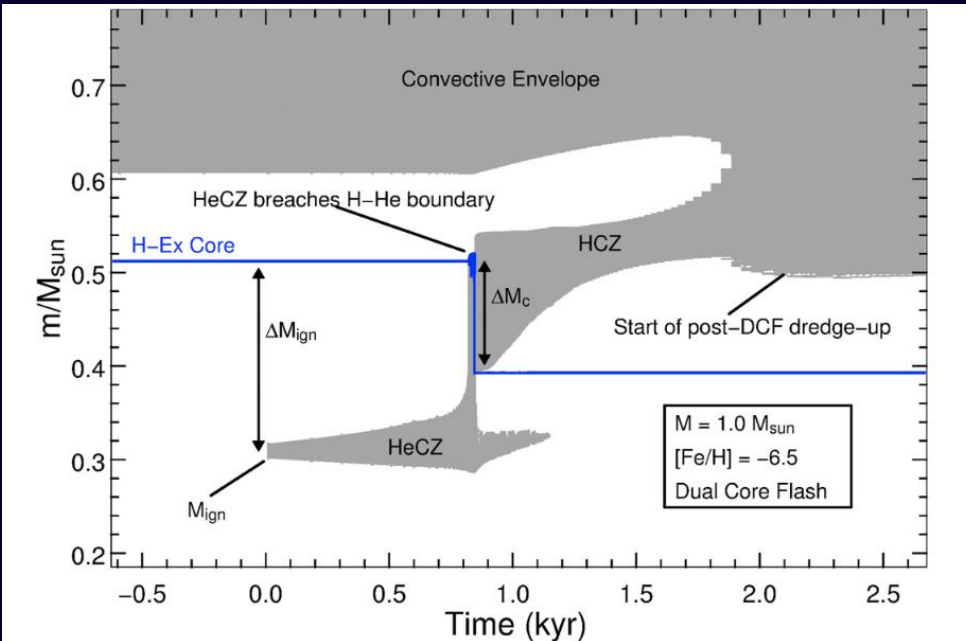


Fig. 1. Example of one of the Dual Core Flash events. Convective regions are shown by grey shading whilst the mass location of the edge of the H-exhausted core is shown by the solid line (blue).

LOW METALLICITY: In $Z_i=0$ and EMP stars, the He-burning convection zone can reach into the H-rich layer, mixing protons into hot He-burning material. Without CNO catalysts, H burning relies on the slower pp chains, producing a cooler, more diffuse H shell and a weaker entropy barrier to He-flash convection.

MULTIPLE SITES: Proton ingestion can occur at different evolutionary stages and in different stellar environments. Two well-known cases are the Dual Core Flash (DCF) during the RGB core-He flash and the Dual Shell Flash (DSF) during the early TP-AGB in intermediate-mass stars.

RUNAWAY FEEDBACK: Ingested protons react with abundant ^{12}C , triggering ^{13}C production and an H-burning flash.

Nucleosynthesis during PIEs

Campbell & Lattanzio (2008, A&A)

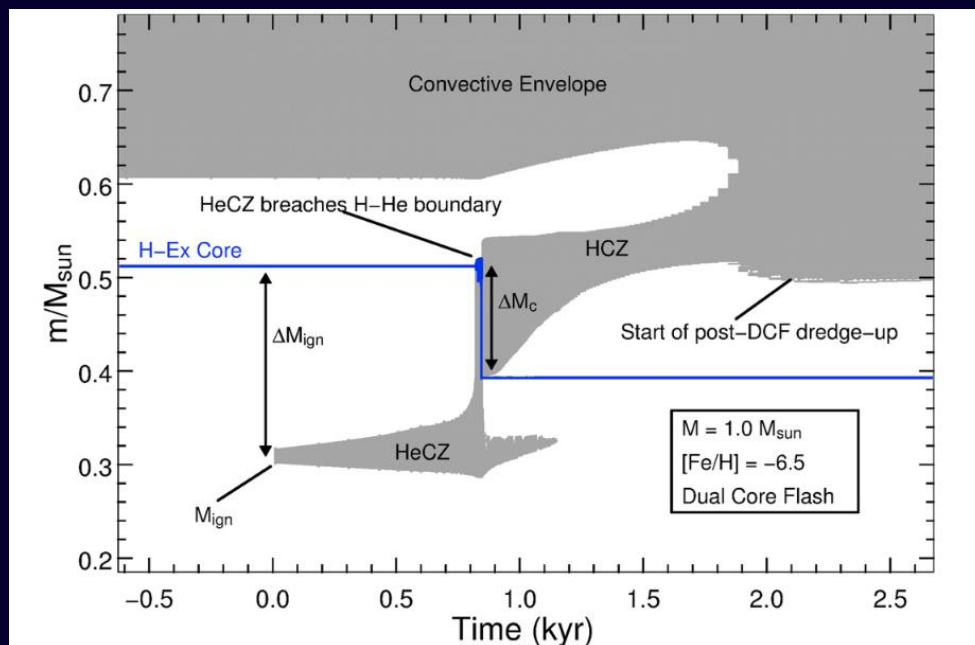


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^{16}O PRODUCTION: $^{12}\text{C}(p,\gamma)^{13}\text{N}(\beta^+)^{13}\text{C}(\alpha,n)^{16}\text{O}$ converts most of the He-flash ^{12}C into ^{16}O . This is a key signature distinguishing PIE pollution from ordinary 3DUP.

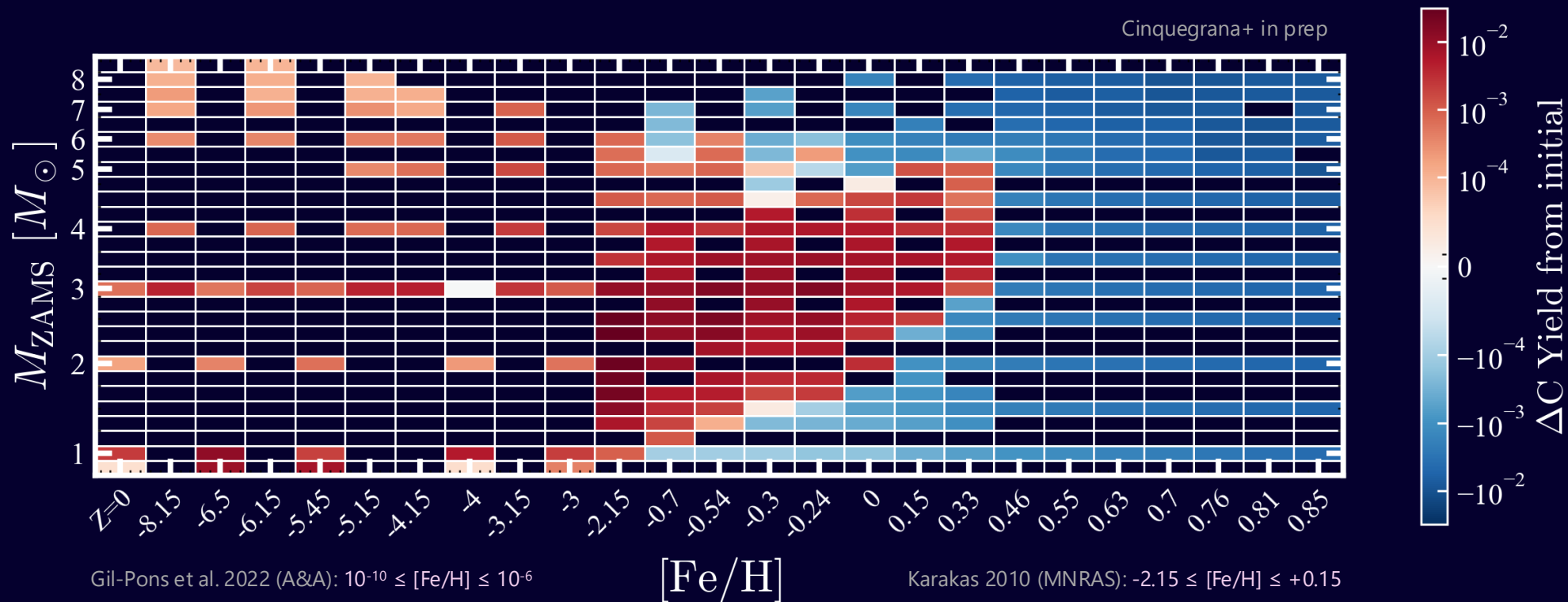
CNO EQUILIBRIUM: A competing channel builds ^{14}N via the CN cycle, leaving diagnostic $^{12}\text{C}/^{13}\text{C}$ ratios as low as $\sim 4-5$.

FREE NEUTRON RELEASE: The same $^{13}\text{C}(\alpha,n)^{16}\text{O}$ reaction liberates neutrons ($Y_n \sim 10^{-13}-10^{-16}$). With ^{14}N still scarce in this metal-poor material, the usual neutron poison is weak.

HEAVY-ELEMENT NUCLEOSYNTHESIS:

Suppressed poisoning allows captures onto heavier nuclei (Sr, Ba, Pb), making PIEs a candidate s-/i-process site in the early universe.

Carbon Yields Across M_{ZAMS} & Z_i



Gil-Pons et al. 2022 (A&A): $10^{-10} \leq [\text{Fe}/\text{H}] \leq 10^{-6}$

Gil-Pons et al. 2021 (A&A): $[\text{Fe}/\text{H}] = -3.15$

Campbell & Lattanzio 2008 (A&A): $Z_i = 0, -6.5 \leq [\text{Fe}/\text{H}] \leq -3$

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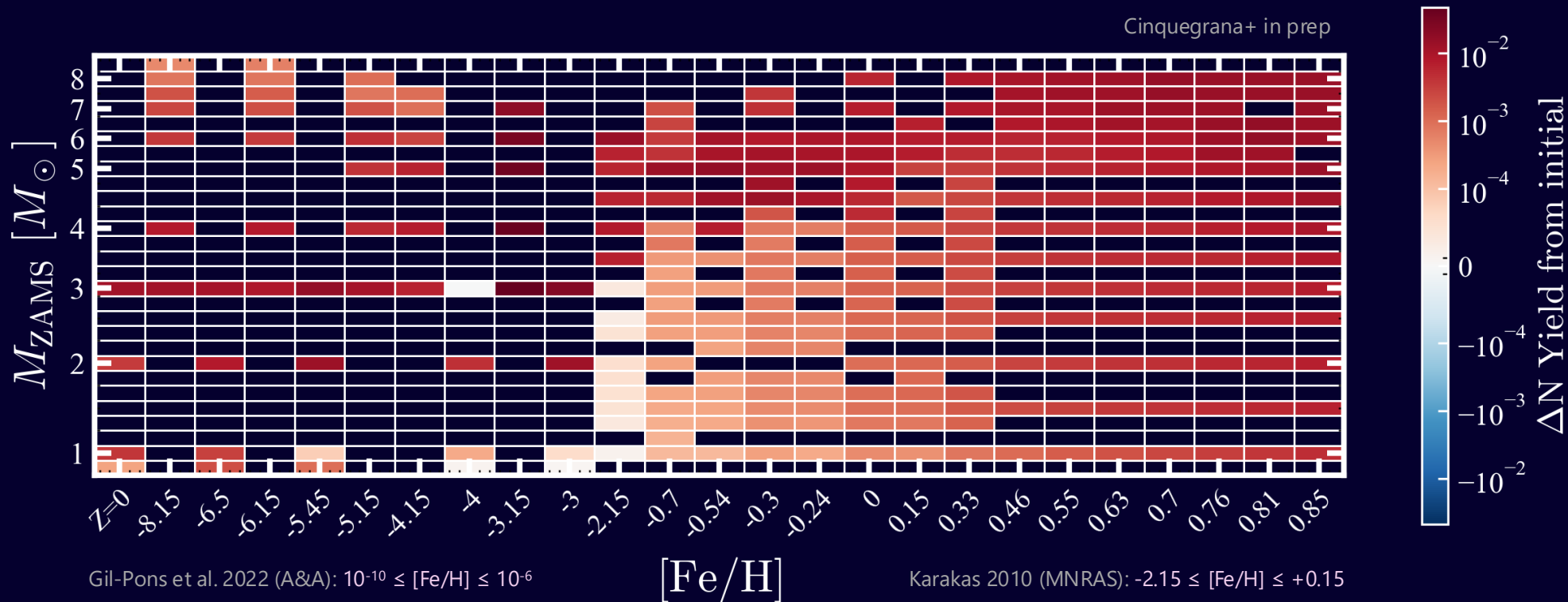
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Final Notes

THE AGB IS A RECYCLING ENGINE: Repeated thermal pulses → third dredge-up → interpulse burning cycle material between the He-intershell and convective envelope, while strong winds return processed material to the ISM.

MASS & METALLICITY SET THE NUCLEOSYNTHETIC REGIME: MZAMS and Z_i determine which burning and mixing processes operate — from third dredge-up and ^{13}C -pocket formation to HBB and the ^{22}Ne neutron source.

NEUTRON SOURCE MATTERS: $^{13}\text{C}(\alpha, n)^{16}\text{O}$ releases a low neutron density over long timescales (interpulse period) whereas $^{22}\text{Ne}(\alpha, n)^{25}\text{Mg}$ is a high neutron density on short timescales (during thermal pulses).

EXTREME CONDITIONS OPEN NEW PATHWAYS: At very low metallicity, proton-ingestion episodes can generate abundant ^{13}C and high neutron fluxes, making the AGB a candidate site for s-/i-process nucleosynthesis and early heavy-element enrichment.

Lab 2 – AGB Nucleosynthesis

REPO: <https://github.com/giulia-cinquegrana/bridgce2026-stellar-nucleosynthesis>

NOTEBOOK: \$REPO/resources/labs/lab2_agb_nucleosynthesis.ipynb

CONTEXT: AGB stars run three neutron sources, each reaching a different neutron density (n_n). Barium's five s-process isotopes are built through a short chain of radioactive caesium branch points, where neutron capture competes with β -decay. Which way each branch tips depends on n_n , so the final Ba isotope pattern is a thermometer / densitometer of the mechanism that made it.

TO DO:

- Run the network - solve the Cs $\rightarrow$ Ba chain at a single n_n and watch the isotopes evolve.
- Score the branch points - compute f_n at the four benchmark densities and work out, by hand, which Ba isotope each stellar mechanism should preferentially feed.
- Scan and connect back - sweep n_n across the full s $\rightarrow$ i-process range, find the two bypass stages in the Ba-ratio plot, then work the discussion questions back to stellar mass and metallicity.

DEFINITIONS:

BRANCH POINT: Nuclide where neutron capture and β -decay compete for the same flow of material.

BRANCHING RATIO (f_n): The fraction of flow through a branch point that proceeds by capture rather than decay.

NEUTRON FLUENCE (τ): Time-integrated neutron exposure.